

Better UTILISATION Of RADIO SPECTRUM With Cognitive Radio

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Cognitive radio is an intelligent technology that improves radio spectrum utilisation. The increasing number of new devices has caused congestion in the ISM (industrial, scientific and medical) radio band. Federal Communications Commission allows unlicensed users (secondary users) to operate in the white space without causing interference to licensed users (primary users). Cognitive radio provides a solution to the additional spectrum requirement for efficient spectrum utilisation. It is capable of sensing and sharing the unused spectrum with unlicensed users.

To enhance the spectrum efficiency, cognitive radio is integrated with different orthogonal frequency-division multiplexing (OFDM) frameworks. Parameters such as probability of detection (P_d) vs probability of false alarm (P_{fa}), P_d vs signal-to-noise ratio (SNR), and bit-error rate (BER) vs SNR for each OFDM framework are calculated and analysed using matched filter detection.

Matched filter method can detect the free portion of primary users' spectrum and allocate it to secondary users, thus minimising spectral overcrowding. A matched filter is obtained by cross-correlating an unknown signal with a known signal to detect the presence of a well-known signal. It is also known as optimal linear filter. In matched filter detection, use of a dynamic threshold improves the spectrum sensing efficiency as well as detection performance at low SNR values.

Integration of cognitive radio with OFDM for better spectrum uti-

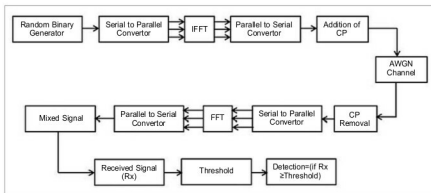


Fig. 1: Block diagram of proposed matched filter detection

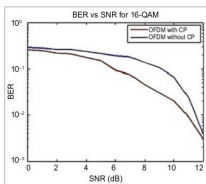


Fig. 2: BER vs SNR for OFDM with and without CP

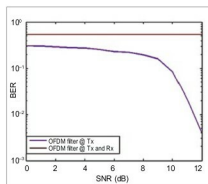


Fig. 3: BER vs SNR for OFDM with filter at transmitter and receiver

lisation requires more priori knowledge of the signal for detection. So, the received signal is demodulated and matched filter applied to the signal. Detection time for matched filter is very less than with other detection methods. Hence, research on integration of cognitive radio with OFDM system is in high demand.

In this article, an OFDM system with and without cyclic prefix (CP), filter at only the transmitter side and filter at both the transmitter and receiver sides, integrated with matched filter detection, is implemented and analysed. The project is implemented using MATLAB version

2013b. It overcomes the requirement of a high SNR in energy detection by using cyclostationary detection. Detection is also faster than with other energy detection schemes.

Matched-filter OFDM model

Block diagram of the proposed matched-filter detection is shown in Fig. 1. A random binary signal is brought forth in a serial way and converted into parallel form. To analyse the signal in time domain, inverse Fast Fourier transform (IFFT) is applied to the signal and it is again converted from parallel into serial form for addition of the